Exit Ticket · reflect and retrieve

Exit Ticket

An embedded OS can never also be real-time.

Use key vocabulary from memory

1. Explain real-time in your own words.

2. Explain embedded in your own words.

3. Explain distributed in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.

Do Now · retrieval

Operating-system types

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which scheduler uses a fixed quantum?

Q2

What does starvation mean?

Q3

What matters most to a real-time task?

Q4

Rising-edge flip-flop Q=0. D pulses high between edges but is 0 at next rising edge. What is new Q?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Real-time OS correctness depends on deadlines, not just average response speed.

Core idea 2

Trace — Embedded and distributed refer to system architecture and purpose; each can overlap with real-time.

Core idea 3

Evaluate — Multi-user concerns people/account isolation, whereas multitasking concerns processes.

Key visual · complete concept

Annotate the key visual

Explain which labels or features show the main process.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why a networked cloud service can be distributed, multi-user and multitasking. [6]

Check that you have used the technical terms and explained your decisions.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Match the five OS categories to an identifying property. [5]

Question 2

Explain OS requirements of a dedicated emergency-stop controller. [6]

Check that you have used the technical terms and explained your decisions.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

real-time — Deadline-bound response.

embedded — Dedicated device software.

distributed — Coordinated networked machines.

multi-user — Account-based sharing.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A safety-critical industrial controller has a fixed response deadline. Which OS type suits it?

Follow the worked method

1. Predict first — Predictable bounded response is required to meet safety deadlines.
2. Model and check — Explain why each value follows from the earlier step.

Source model · final illustrated step

IDO · STEP 2

Predictable bounded response is required to meet safety deadlines.

Explain why each value follows from the earlier step.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Many accounts require permissions and several running processes share CPU time.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.